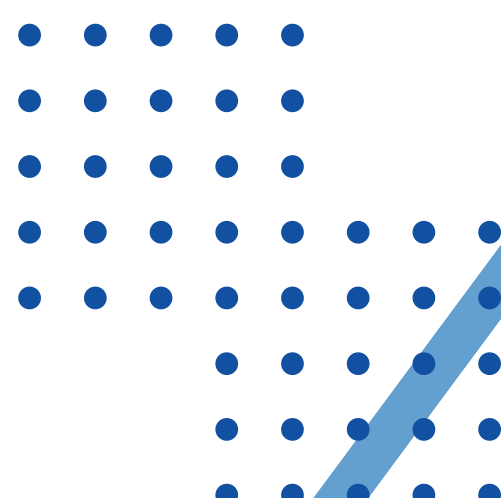




Smart City Planning using EarthVQA



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Background

- Urban planning requires analyzing complex spatial data such as buildings, roads, vegetation, and water bodies. Traditional methods rely heavily on manual inspection of satellite images, which is time-consuming and prone to human error.
- With advancements in machine learning and computer vision, automated systems can analyze satellite imagery, extract spatial features, and provide intelligent insights. This enables faster, scalable, and data-driven decision-making for smart city development.

Problem Statement and Objective

Problem Statement:

- Urban planners lack automated tools to assess land suitability, infrastructure quality, and environmental risks from satellite data.

Our objectives are:

- Automated Satellite Image Segmentation
- Visual Question Answering (VQA) on spatial data
- Smart City Planning Recommendations
- Flood Risk, Density, and Infrastructure Analysis

Project Overview

Purpose:

- Analyze satellite images and generate intelligent urban planning insights.

Key Features:

- Semantic segmentation of satellite images
- Spatial feature extraction (buildings, roads, water, vegetation)
- Decision engine for planning recommendations
- Question-answering system for user queries

Input & Output Pipeline

Input:

- Satellite Image (512×512 PNG)
- Natural Language Question

Output:

- Segmentation Mask (8 classes)
- Spatial Metrics (16 features)
- Planning Report (4 scores)
- Intelligent Answers + Recommendations

8 Segmentation Classes:

0 — Background

1 — Building

2 — Road

3 — Water

4 — Barren

5 — Forest

6 — Agriculture

7 — Playground

Dataset Overview

Source: EarthVQA Dataset
Details:

- 2,522 training images
- 1,809 test images
- 63,000+ QA pairs

Image Specs:

- Resolution: 512×512
- RGB satellite images

Supported intent types:

— counting	→ "How many buildings..."	→ count from spatial features
— judging	→ "Are there any roads..."	→ yes/no from features
— density	→ "Is this overcrowded..."	→ density score report
— risk	→ "Is there flood risk..."	→ flood risk analysis
— situation	→ "How is the road..."	→ infrastructure report
— relation	→ "Are buildings near water..."	→ proximity analysis
— planning	→ "Is this area suitable..."	→ full planning report

Supported intent types

Methodology

Step 1: Semantic Segmentation (SemanticFPN)

- Input: Satellite Image (512×512 RGB)
- Backbone: ResNet50 for feature extraction
- Feature Pyramid Network (FPN) for multi-scale representation
- Pixel-wise classification into 8 land-use classes

Output:

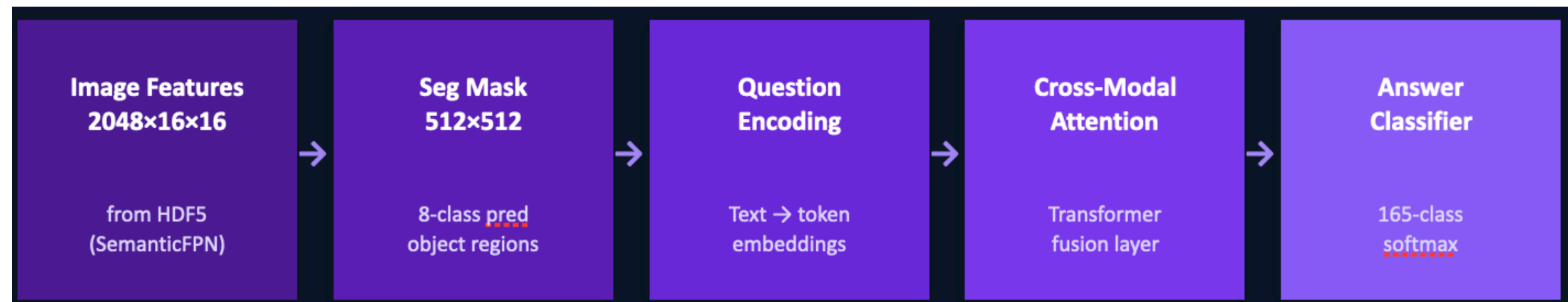
- Segmentation Mask (512×512)
- Deep Feature Tensor (2048×16×16)

Step 2: Visual Question Answering (SOBA Model)

- Input: Image features + Natural language question
- Transformer-based attention mechanism
- Spatial Object-Based Attention for region focus
- Classification over 165 answer categories

Output:

- Context-aware predicted answers



Methodology

Step 3: Spatial Feature Extraction (Spatial Engine)

- Extract quantitative features from segmentation mask:
 - Building count, road density, vegetation %, water area

Step 4: Decision Engine (Planning Intelligence)

- Compute 4 planning scores:
 - Density Score
 - Green Coverage Score
 - Flood Risk Score
 - Infrastructure Score
- Weighted scoring and threshold-based classification
- Generate planning recommendations

```
4-Score Assessment:
├─ 🏠 Density Score (0-1)
│  └─ Weighted:  $0.4 \times \text{building\_count\_norm} + 0.6 \times \text{built\_area\_pct}$ 
│     └─ Thresholds: low < 0.07, moderate < 0.34, high > 0.70
├─ 🌳 Green Coverage Score (0-1)
│  └─ vegetation_area_pct normalized against target (36%)
│     └─ Thresholds: insufficient < 10%, adequate ≥ 36%, good ≥ 66%
├─ 🌊 Flood Risk Score (0-1)
│  └─ Weighted:  $0.5 \times \text{water\_proximity} + 0.3 \times \text{water\_area} + 0.2 \times \text{ratio}$ 
│     └─ Safe distance: 50px (~50m). Below = high risk
├─ 🛣️ Infrastructure Score (0-1)
│  └─ Weighted:  $0.5 \times \text{road\_coverage} + 0.3 \times \text{intersections} + 0.2 \times \text{connectivity}$ 
│     └─ Thresholds: poor < 0.11, moderate < 0.32, good > 0.70
└─ Overall Suitability = weighted average of 4 scores
   Labels: Not Suitable | Needs Improvement | Moderately Suitable | Suitable | Highly Suitable
```

Step 5: Question Answering & Pipeline Integration

- Intent parsing of user query (counting, risk, planning, etc.)
- Combine:
 - Spatial features
 - Decision scores
 - VQA predictions
- Generate final enriched answer with insights

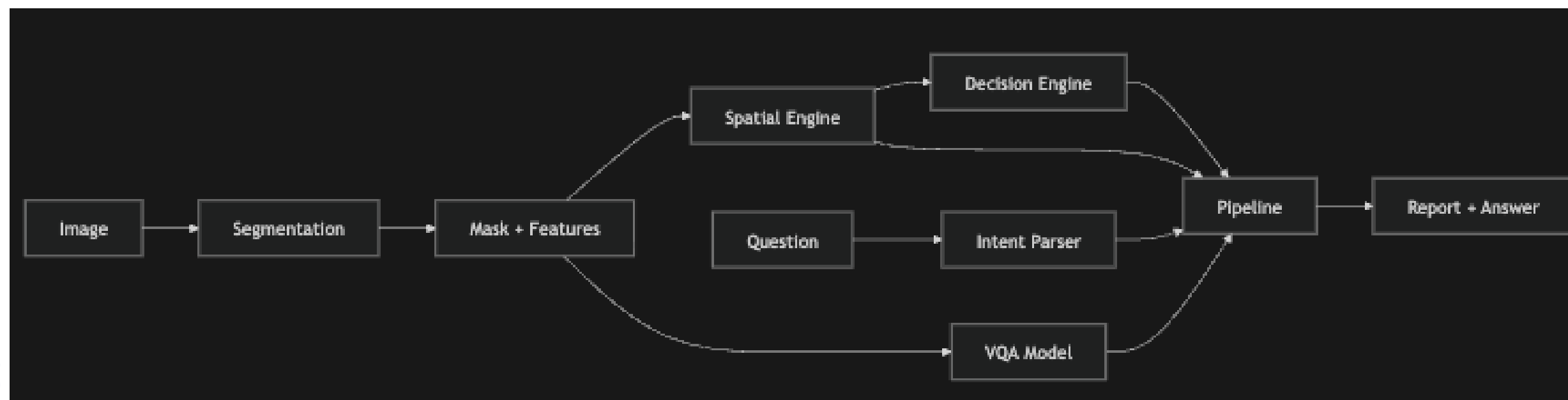
Complete Pipeline

Complete Flow:

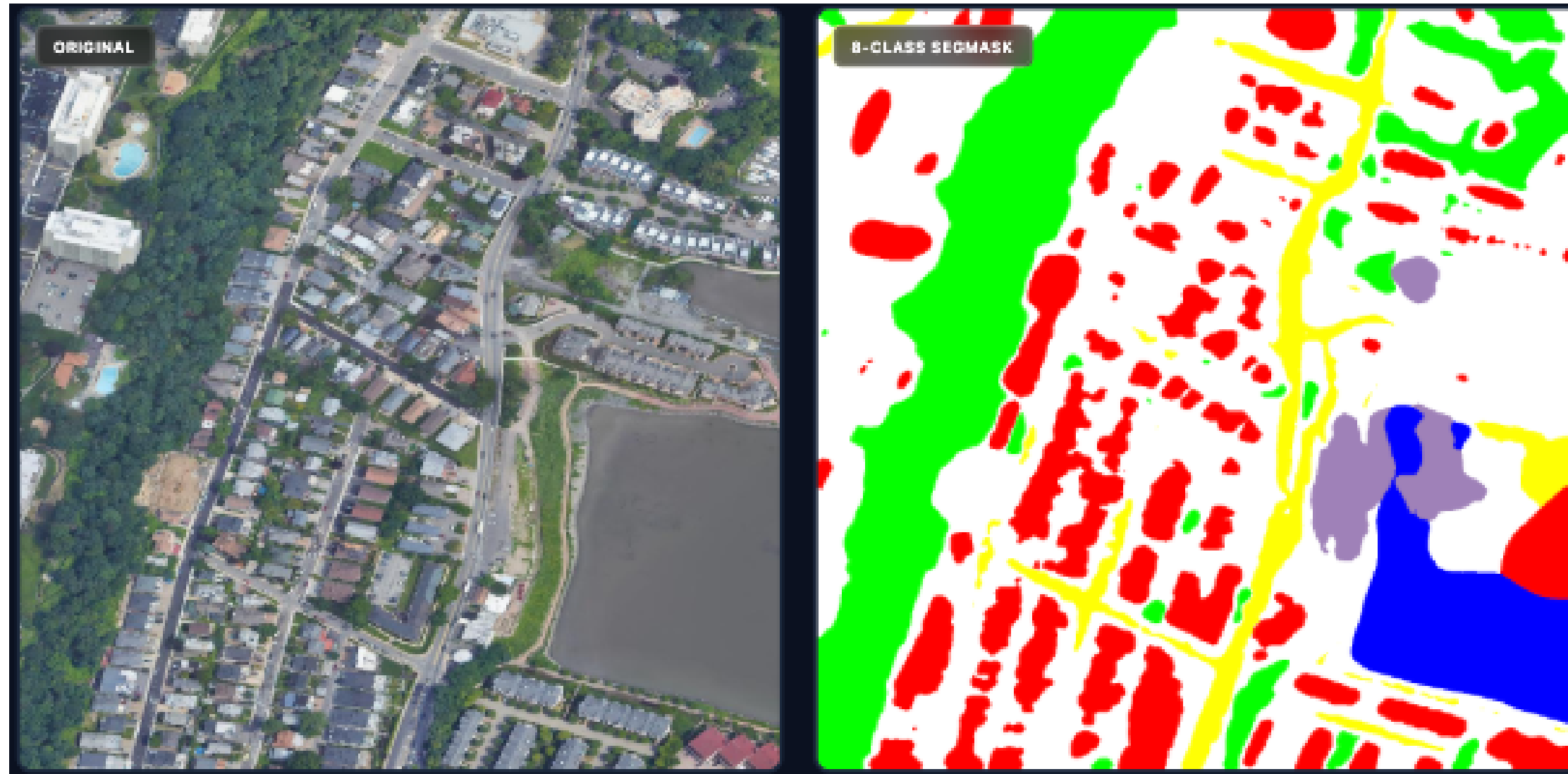
- Input: Satellite Image + User Question
- Semantic Segmentation (SemanticFPN) → 8-class Mask + Features
- VQA Model (SOBA) → Answer Predictions
- Spatial Engine → Extract 16 spatial metrics
- Decision Engine → Compute 4 planning scores
- Intent Parser → Understand user query
- Unified Pipeline → Combine ML + Spatial reasoning

Output:

- Planning Report (Density, Green, Flood, Infrastructure)
- Intelligent Answer with recommendations



Results



Sample image tested and the segmented mask is generated for it

Results

Table 2: Semantic segmentation results obtained on the Test set of LoveDA.

Method	Backbone	IoU per category (%)							mIoU (%)
		Background	Building	Road	Water	Barren	Forest	Agriculture	
FCN8S [19]	VGG16	42.60	49.51	48.05	73.09	11.84	43.49	58.30	46.69
DeepLabV3+ [5]	ResNet50	42.97	50.88	52.02	74.36	10.40	44.21	58.53	47.62
PAN [17]	ResNet50	43.04	51.34	50.93	74.77	10.03	42.19	57.65	47.13
UNet [32]	ResNet50	43.06	52.74	52.78	73.08	10.33	43.05	59.87	47.84
UNet++ [55]	ResNet50	42.85	52.58	52.82	74.51	11.42	44.42	58.80	48.20
Semantic-FPN [16]	ResNet50	42.93	51.53	53.43	74.67	11.21	44.62	58.68	48.15
PSPNet [53]	ResNet50	44.40	52.13	53.52	76.50	9.73	44.07	57.85	48.31
LinkNet [3]	ResNet50	43.61	52.07	52.53	76.85	12.16	45.05	57.25	48.50
FarSeg [54]	ResNet50	43.09	51.48	53.85	76.61	9.78	43.33	58.90	48.15
FactSeg [23]	ResNet50	42.60	53.63	52.79	76.94	16.20	42.92	57.50	48.94
HRNet [45]	W32	44.61	55.34	57.42	73.96	11.07	45.25	60.88	49.79

Method	Promp.	↑Accuracy(%)						↑OA(%)	↓RMSE		↓OR
		Bas Ju	Rel Ju	Bas Co	Rel Co	Obj An	Com An		Bas Co	Rel Co	
SAN	×	87.59	81.79	76.26	59.23	55.00	43.25	75.66	1.136	1.318	1.160
MAC	×	82.89	79.46	72.53	55.86	46.32	40.50	71.98	1.407	1.337	1.398
BUTD	✓	90.01	82.02	77.16	60.95	56.29	42.29	76.49	0.890	1.292	0.950
BAN	✓	89.81	81.87	77.58	63.71	55.67	45.06	76.74	0.819	1.241	0.883
MCAN	✓	89.65	81.65	79.83	63.16	57.28	43.71	77.07	0.816	1.230	0.879
D-VQA	✓	89.73	82.12	77.38	63.99	55.14	43.20	76.59	0.916	1.238	0.962
BLIP-2	×	88.13	81.92	70.26	58.58	42.72	28.34	71.07	1.879	1.320	1.818
Instruct-BLIP	×	89.67	79.69	76.96	63.34	59.72	45.68	75.25	0.799	1.217	0.862
RSVQA	×	82.43	79.34	70.68	55.53	42.45	35.46	70.70	1.733	1.359	1.691
RSIVQA	×	85.32	80.44	75.01	56.63	51.55	39.25	73.71	1.718	1.346	1.676
SOBA (ours)	✓	89.63	82.64	80.17	67.86	61.40	49.30	78.14	0.785	1.145	0.839

Table 1: Compared results with other VQA methods on EarthVQA^{test}

The ResNet-50 SFPN achieves roughly 48.15% mIoU (which is an incredibly strong result for an 8-class remote sensing problem due to the extreme variation in satellite scale)

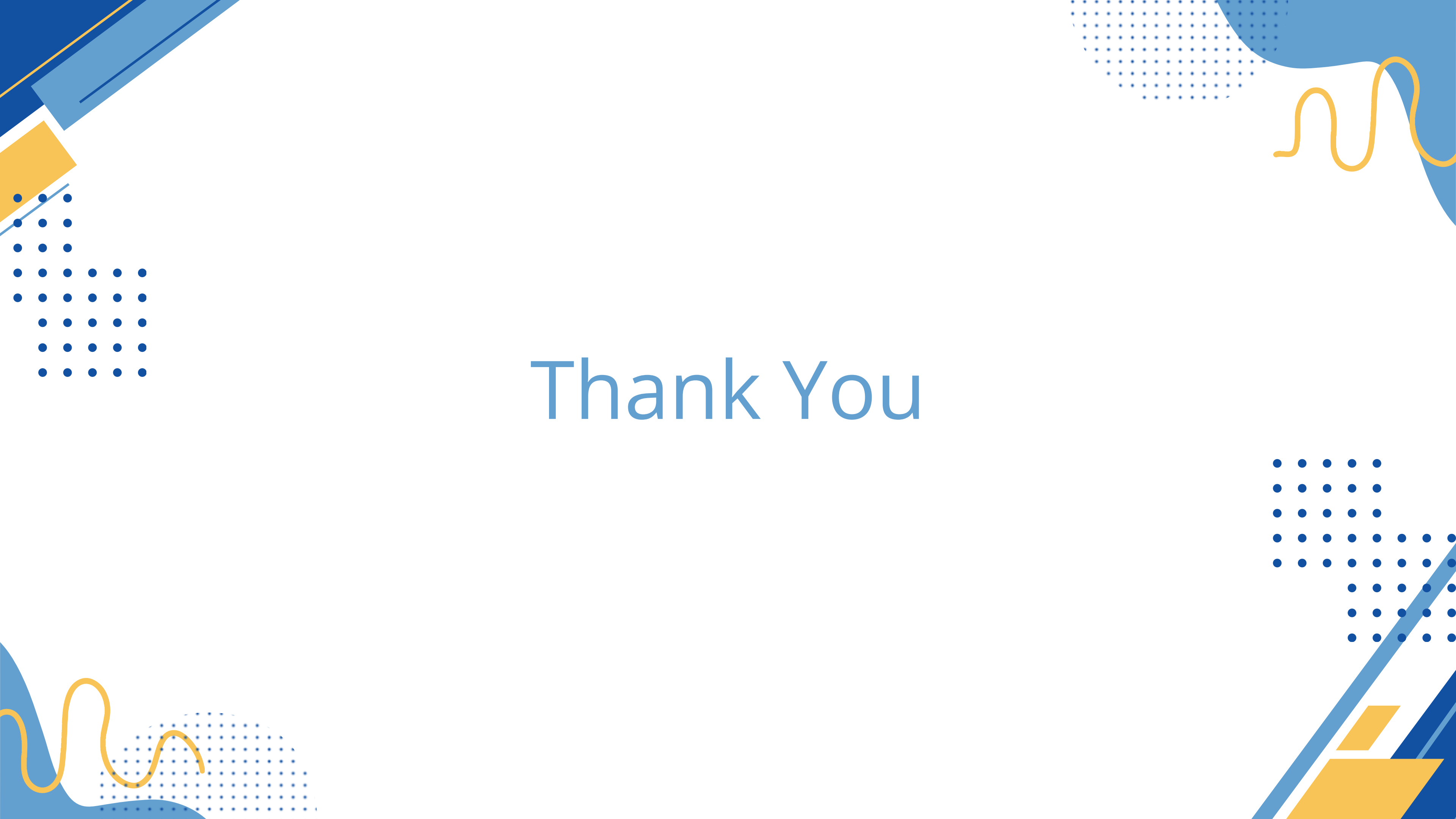
The SOBA architecture sits at 78.4% Accuracy on the complex relational questions inside the EarthVQA validation set.

Future Work

- Develop a complete frontend dashboard (React-based UI) for visualization
- Introduce a chat-based interface for user interaction Users can ask questions in natural language (e.g., “Is this area suitable for development?”)
- Urban Heat Island & Air Quality Prediction: Use spatial features like building density, vegetation, water, and roads as inputs to ML models such as XGBoost or LightGBM. Predict environmental factors like land surface temperature and air quality to assess the impact of urban structures.

Future Work

- Time-Series Change Detection (Urban Sprawl):
Analyze satellite images of the same region over time and compute changes in spatial features such as vegetation and buildings. Train models like SVM or Random Forest to classify patterns like deforestation, urban expansion, or sustainable growth.

The slide features decorative geometric patterns in the corners. The top-left corner has a blue triangle with a yellow rectangle and a grid of blue dots. The top-right corner has a blue shape with a yellow wavy line and a grid of blue dots. The bottom-left corner has a blue shape with a yellow wavy line and a grid of blue dots. The bottom-right corner has a blue triangle with a yellow rectangle and a grid of blue dots.

Thank You